30054 International and Monetary Economics

Spring 2020

Exam 1 (May 20, 2020) Solution

1. Gross national expenditure (GNE) is

   (a) GDP plus net factor income from abroad (NIFA)

   (b) GDP plus net unilateral transfers (NUT)

   (c) GDP minus trade balance (NX)

   (d) GDP minus consumption (C)

2. A Japanese resident purchases $100 million of US Treasury bonds from a US resident. After this transaction, Japanese financial account balance defined as in our class (i.e., so that FA+KA+CA=0)
   Note that this question was designed to have only one correct answer. However, another correct answer unintentionally slipped through. FA declines if the investor pays with domestically owned Yens in exchange for US bonds, and a US resident uses these Yens to buy a Japanese robot (increasing Japanese CA). If, instead, the investor pays with Japanese bonds or if he holds a US deposit account and uses it to pay for the US bonds, then there is an offsetting FA transaction.

   (a) increases;

   (b) declines;

   (c) does not change;

   (d) None of the above.

3. The UIP states that domestic nominal interest rate equals foreign nominal interest rate

   (a) plus the expected rate of depreciation of domestic currency;

   (b) minus the expected rate of depreciation of domestic currency;

   (c) plus the expected rate of appreciation of domestic currency;

   (d) plus the expected rate of depreciation of foreign currency;

4. The AA schedule describes equilibrium on ...

   (a) candy market;

   (b) goods market;

   (c) foreign exchange market and money market;

   (d) None of the above.

5. When (non-US) domestic producers set their export prices in US dollars when they export their goods to countries other than the US, we call this

   (a) producer-currency pricing (PCP);
   (b) local-currency pricing (LCP);
   (c) dominant currency paradigm (DCP);
   (d) gross domestic product (GDP).

6. The absolute PPP theory of the nominal exchange rate determination implies that the real exchange rate equals

   (a) one;
   (b) two;
   (c) three;
   (d) four.

7. The doom loop that we covered in class

   (a) proposes that self-fulfilling expectations can ruin both the financial and public sectors of a country;
   (b) is the name of a release of the video game Doom;
   (c) is statistical software that can be used to estimate regressions;
   (d) is the name of the stimulus program that the ECB launched to fight the effects of the coronavirus.

8. The theory of Optimum Currency Areas proposes that it makes more sense to establish a currency union when

   (a) countries labor markets are integrated;
   (b) countries goods markets are integrated;
   (c) countries experience symmetric shocks;
   (d) All of the above.

9. Which of the following groups of models that we discussed in class emphasize the importance of current or future fundamental economic conditions as opposed to the multiple equilibria situation in explaining balance-of-payments crises?

   (a) The 1st and 2nd generation models;
   (b) The 2nd and 3rd generation models;
   (c) The 1st and 3rd generation models;
   (d) None of the above.

10. Under The Bretton Woods System, all but one participating countries formally fixed their exchange rates to

    (a) The Italian Lira;

    (b) The British Pound;

    (c) The Soviet rubble;

    (d) None of the above.

11. Indicate if a statement is true or false. Provide a concise justification.
"A situation in which a country has a fixed exchange rate and an independent monetary policy is impossible."

    False. For example, the country can fix the exchange rate and have an independent monetary policy when it limits the cross-border capital mobility.

12. Indicate if a statement is true or false. Provide a concise justification. "Floating exchange rate regimes are superior to fixed exchange rate regimes."

    False. In general, it is impossible to argue that one regime dominates the other. Both regimes have pros and cons. For example, one disadvantage of a floating exchange rate regime is that it is harder to make future international trade plans when the exchange rate is very volatile.

13. Indicate if a statement is true or false. Provide a concise justification. "Eurozone is an optimum currency area."

    False (but read the last sentence of the answer). While the goods markets are quite integrated, the labor markets are country-specific in the Eurozone. Moreover, shocks are quite asymmetric across the Euro area. As a result, the Eurozone does not look like the optimum currency area in the sense of Mundell (1961). However, I emphasized in class that it is quite challenging to measure how far the Eurozone is from the optimum currency area.

14. Indicate if a statement is true or false. Provide a concise justification. "Expectations about the future do not matter for the exchange rate determination."

    False. Consider the UIP equation studied in class. It features expectations of future exchange rate.

15. Indicate if a statement is true or false. Provide a concise justification. "If the capital account of a country is zero ($KA = 0$), then a country can be a net capital receiver ($FA > 0$) and run positive current account surplus ($CA > 0$)."

    False (but read the last sentence). If this was true, then the accounting identify $FA + CA + KA = 0$ would be violated. However, when we deal with real world data, FA and CA can be mismeasured. In such a case, it is possible that $FA + CA + KA \neq 0$.

16. Which of the following formulas represent the DD schedule discussed in class? (The notation is the same that we used in class)

    (a) $\overline{M}/\overline{P} = L[\overline{R}^* + (E^e - E)/E, Y]$ or, equivalently, $Y = Y(E, E^e, \overline{R}^*, P, \overline{M})$;

    (b) $Y = C(Y - \overline{T}) + \overline{I} + \overline{G} + NX\left(E\overline{P}^*/\overline{P}, Y - \overline{T}\right)$;

    (c) $R = \overline{R}^* + (E^e - E)/E$;

    (d) $FA + CA + KA = 0$.

17. Under the assumptions used in class (consumption responds less than one-to-one to disposable income and net exports increase in the real exchange rate), how does the DD schedule look like in the exchange rate/output space? Choose one correct answer from Figure 17.

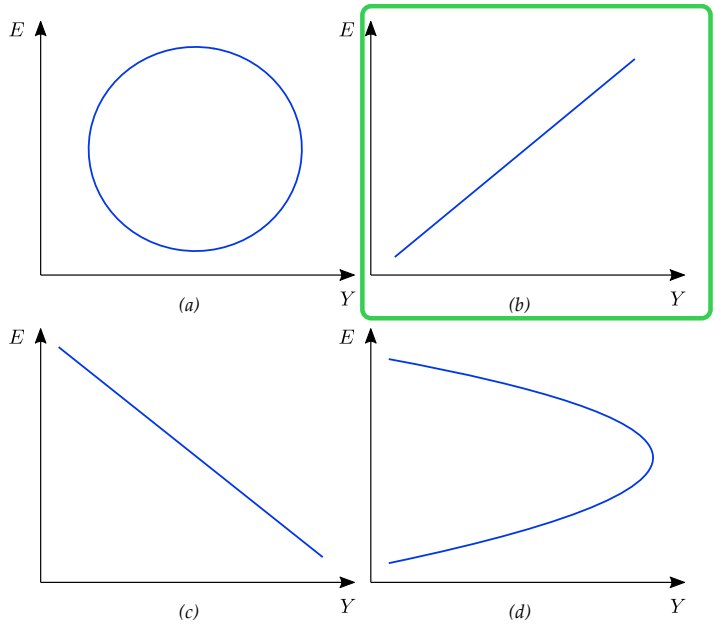

Figure 1: The DD schedule.

18. Which of the following formulas represent the AA schedule discussed in class? (The notation is the same that we used in class)

    (a) $\overline{M}/\overline{P} = L[\overline{R}^* + (\overline{E}^e - E)/E, Y]$ or, equivalently, $Y = Y(E, E^e, \overline{R}^*, P, \overline{M})$;

    (b) $Y = C(Y - \overline{T}) + \overline{I} + \overline{G} + NX\left(E\overline{P}^*/\overline{P}, Y - \overline{T}\right)$;

    (c) $R = \overline{R}^* + (E^e - E)/E$;

    (d) $FA + CA + KA = 0$.

19. Which of the scenarios depicted in the figure below represents a response of the AA curve to a temporary expansion of the supply of money?

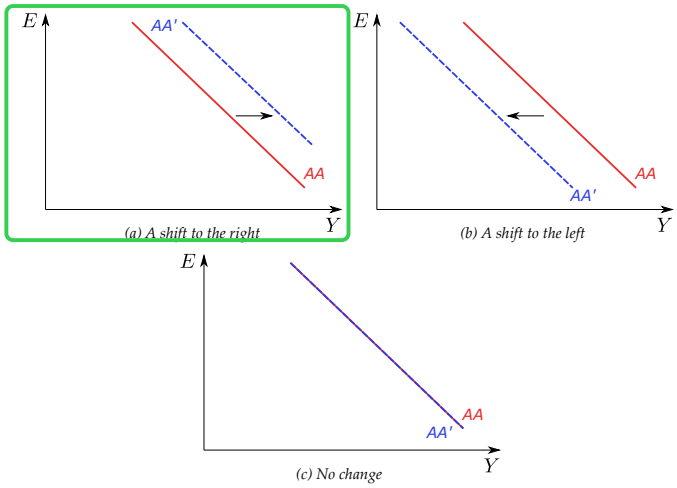

Figure 2: The AA schedule change after a temporary monetary policy expansion.

20. The figure below shows the AA schedule in the normal and the liquidity trap periods. Explain briefly why the AA curve becomes flat in the liquidity trap.

    In the liquidity trap, the interest rate is stuck at zero. Using the UIP condition, we get that exchange rate is only influenced by the future exchange rate and foreign interest rate. Home interest (and output that affects it) does not affect the exchange rate in liquidity trap in our model.

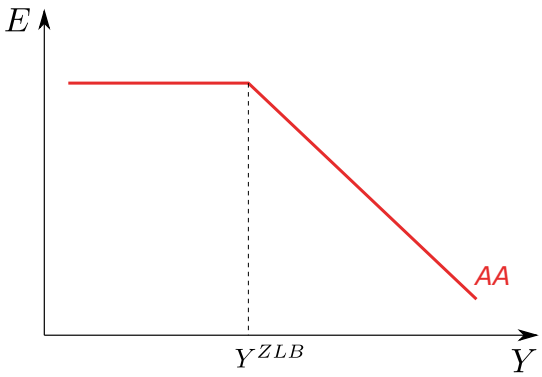

Figure 3: The AA schedule under liquidity trap and normal conditions.

21. Which of the scenarios depicted in Figure below represents a response of the AA schedule to a temporary expansion of the supply of money?

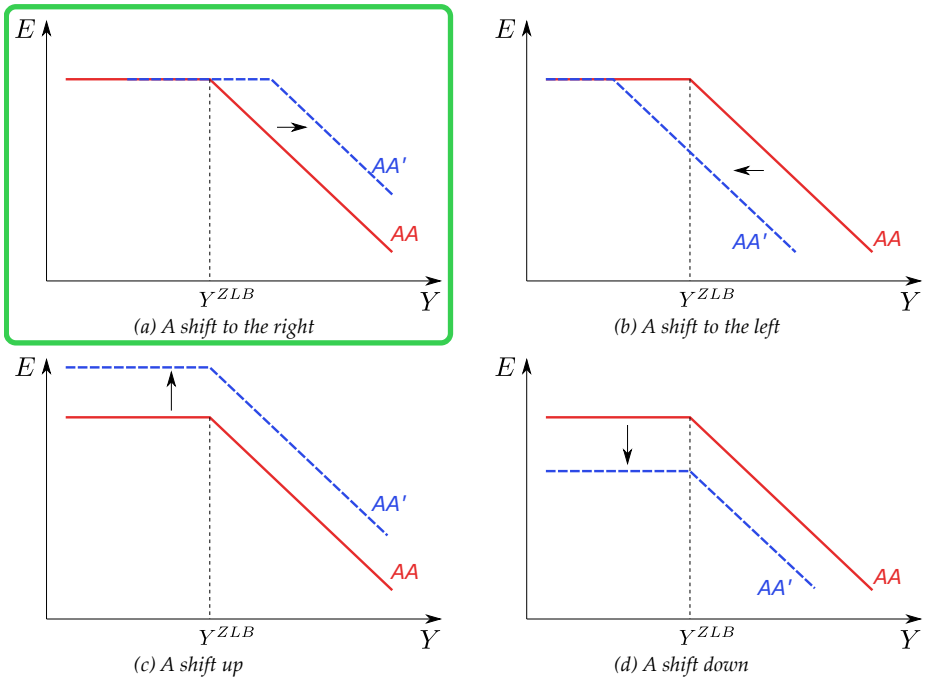

Figure 4: The AA schedule change after a temporary monetary policy expansion.

22. Use your answer to the previous question 21 and the assumption that the DD curve intersects the AA curve in its flat part, as depicted in the figure below. A (small) temporary expansion of money supply

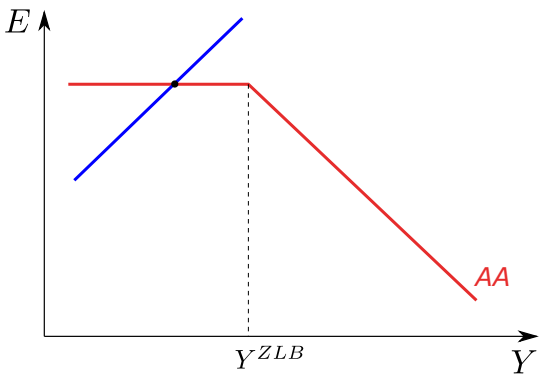

Figure 5: The AA and DD schedules in liquidity trap.

(a) increases current output;
(b) decreases current output;
(c) does not change current output;
(d) all of the above.

23. Which of the scenarios depicted in the figure below represents a response of the AA schedule to the central bank's promise to increase the supply of money permanently starting from the next year (a form of forward guidance policy)? Assume that market participants believe in the central bank's promise.

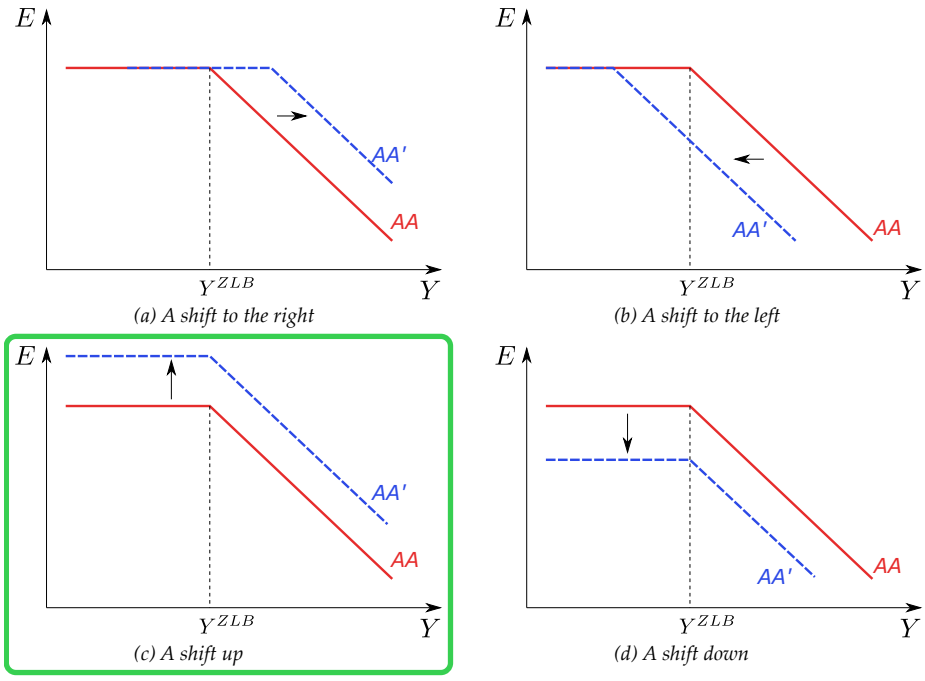

Figure 6: The AA schedule change after a promise of a future permanent monetary policy expansion.

24. Use your answer to the previous question 23 and the assumption that the DD curve intersects the AA curve in its flat part, as depicted in the figure below. A central bank's credible promise to permanently increase the supply of money in the next year by a small amount

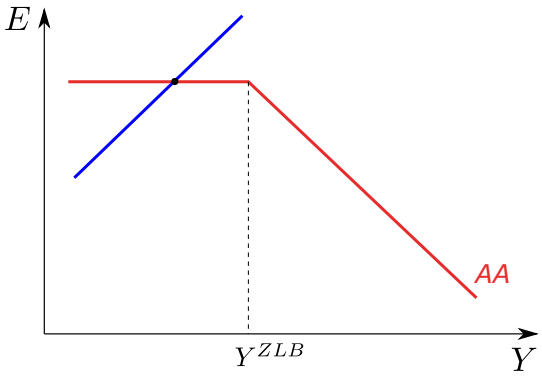

Figure 7: The AA and DD schedules in liquidity trap.

(a) increases current output;

(b) decreases current output;

(c) does not change current output;

(d) all of the above.